MIL-E-1/1706
29 April 1971
SUPERSEDING
MIL-E-1/268B
26 June 1970

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5998A

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Twin triode, low Mu

Outline	---	12-15 (EIA)
Base	---	B8-110 (phenolic, short medium shell)
Envelope	---	T12
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	2g	2a	2k	1g	1a	1k	h	h

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Rk/k	Rg/g	Ib/a	Pp/a	Ehk	TE	Alt
Unit:	V	Vdc	Vdc	Ohms	Meg	mAdc	W	v	°C	ft
Maximum:	6.9	275	---	---	0.25	140	15	100	175	Note 1
Minimum:	5.7	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	110	0	105	---	---	---	---	---	---

GENERAL:

Qualification - Not required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 1							
1301	Heater current		0.65	II	If	2.20	2.60	A
1266	Total grid current	Ec = -14 Vdc; Eb = 120 Vdc; Rg = 0.1 Meg; Rk = 0 (see notes 2, 3, and 4)	0.65	II	Ic	0	-3.0	µAdc
1306	Transconductance (1)	See note 5	0.65	II	Sm	12,500	18,500	µmhos
1306	Transconductance (2)	Ef = 5.7 V (see note 5)	0.65	II	Sm Ef	11,500	---	µmhos
1336	Heater-cathode leakage	Ehk = 200 Vdc (see note 2)	0.65	II	Ihk	---	60	µAdc
1256	Electrode current (1) (anode)	See note 5	0.65	II	Ib	75	125	mAdc
1256	Electrode current (2) (anode)	Eb = 60 Vdc; Ec = 0; Rk/k = 0 (see notes 4 and 5)	0.65	II	Ib	135	---	mAdc
1256	Electrode current (3) (anode)	Eb = 200 Vdc; Rk/k = 0; Ec = -55 Vdc (see note 5)	0.65	II	Ib	---	10	mAdc
1201	Short and discontinuity detection		0.40	II	---	---	---	---
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	Ebb = 120 Vdc; Rp = 2,000 ohms; Ec/Ib = 45 mAdc (see note 2)	---	---	Ep	---	200	mVac
1316	Amplification factor	See note 5	---	---	Mu	4.1	6.7	---
1211	Insulation of electrodes		2.5	I	R	100	---	Meg
1111	Base pin solder depth		---	---	---	---	---	---
1101	Secureness of base, cap, or insert		---	---	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 3							
1501	Intermittent life	Group A; Ef = 6.3 V; Eb = 125 Vdc; Rk/k = 110 ohms; Rg/g = 0.25 Meg; Ehk = 100 Vdc; TE = 175°C (min) (see note 6)	---	---	---	---	---	---
---	Intermittent life-test end points (500 hours)	Electrode current (2) (anode)	---	---	Ib	125	---	mAdc
		Total grid current	---	---	Ic	---	-6.0	μAdc
		Transconductance (1)	---	---	Sm	11,500	---	μmhos
		Insulation of electrodes	---	---	R	25	---	Meg

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. Tie 1k to 2k; 1g to 2g; and 1a to 2a.
3. Approved alternate test. The insertion of 0.27 Meg resistor in the grid circuit shall not change the anode current more than 7.0 mAdc.
4. This test to be performed at the conclusion of the holding period.
5. Apply voltages to both sections but test each section separately.
6. *Envelope temperature* (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MU
Air Force -
DSA - ES

User activities:
Army - WC
Navy - AS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

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